

RegimeGate

An Adaptive Fusion Controller for Non-Stationary Demand

Detailed Solution & Prototype Document ■ Round 2

Problem Statement 3 — Innovation Submission ■ Team **Absolute** ■ Team Lead: Ayan Ahmed Khan

Executive summary. Demand in modern supply chains is non-stationary: no single forecaster is best across calm and turbulent regimes. The standard remedy — a fixed weighted blend or a one-shot stacked meta-learner — assumes the relative competence of each model never changes, which is precisely wrong. **RegimeGate** is a tiny ($< 50k$ -parameter) context-conditioned gating network that learns *where* each forecaster should be trusted and emits dynamic fusion weights over two frozen experts — LightGBM (tree) and N-HiTS (deep) — conditioned on the *observable demand regime*, not on the predictions. That distinction is what places it provably beyond a stacked meta-learner.

We validate the full system on the real Walmart **M5** benchmark with a leakage-safe, rolling-origin protocol. The headline results: RegimeGate achieves the **best overall WRMSSE**, **beats the predictions-only stacker by $\sim 15\%$** , recovers a genuine two-sided regime split (N-HiTS wins smooth demand, LightGBM wins intermittent), exposes an interpretable, tunable **accuracy–robustness dial**, and degrades gracefully under a calibrated shock battery — all at negligible added compute, on a free Colab T4 GPU.

Headline (real M5)	Result
Overall WRMSSE — RegimeGate vs. stacked meta-learner	+15.5% better
Overall WRMSSE — best of all 5 variants	0.734 (wins)
Two-sided regime split	N-HiTS wins smooth; LightGBM wins intermittent
Shock robustness (robustness dial)	matches the most-robust expert on all 3 shocks
Gate size / added compute	610 params; negligible

1 Problem Understanding

Demand is **non-stationary**: its statistical behaviour shifts continuously between calm, seasonal stretches and abrupt shocks driven by promotions, stock-outs, weather, and macro disruptions such as the COVID-19 surge or the Suez Canal blockage. Large-scale evidence from the M5 (Walmart) competition shows that gradient-boosted trees (LightGBM) dominate on granular, intermittent, volatile series, while deep sequential models (N-HiTS, N-BEATS, TFT) are most competitive on smoother, higher-level demand. Practitioners therefore combine both families — but almost always through a **fixed weighted average** (e.g. a static 60/40 blend) or, at best, a **stacked meta-learner** trained once on the model outputs.

Why fixed fusion is mismatched

A constant weight assumes each model’s relative competence never changes — yet the entire premise of combining a deep model with a tree ensemble is that their competence is **regime-dependent**. A weight that is optimal during a stable season is provably sub-optimal during a demand shock, and vice-versa. The cost is concrete: every avoidable forecast error becomes either overstock (tied-up

capital, waste) or stockout (lost sales, eroded service).

The sharpened, falsifiable problem

Adversarial review (we used Claude as a sceptical judge) reshaped a vague goal into a testable one. A stacked meta-learner taking only the two forecasts learns a single fixed blending rule of the outputs; two samples with *identical* base predictions but in *different* volatility regimes would receive identical weights — the behaviour we actually need is inexpressible by a pure stacker. The problem therefore has two pillars:

- **Condition on exogenous context.** Fusion weights must depend on observable descriptors of the regime (volatility, intermittency, seasonality, shock proximity), not on the component predictions alone.
- **Define the regime rigorously and leakage-safely.** We use the Syntetos–Boylan ADI/CV² taxonomy computed only on past windows — a continuous, scale-free signal — rather than a brittle hard threshold.

The right success question is not whether WRMSSE is lower in the abstract, but whether adaptivity reduces the **costliest** errors — stockouts during shocks, overstock during lulls — without adding operational risk or material compute, under a leakage-safe protocol with an interpretability and a robustness obligation.

2 Solution Architecture

RegimeGate is a thin, model-agnostic **Adaptive Fusion Controller** that sits on top of any set of trained forecasters. It has three parts.

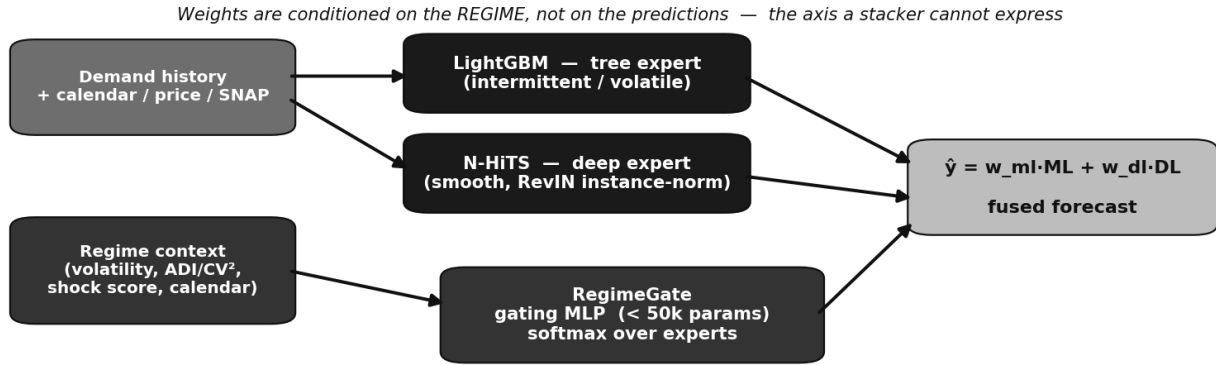


Figure 1: RegimeGate: two frozen experts forecast in parallel; a context-conditioned gating network emits the fusion weights, conditioned on the regime rather than on the predictions.

Two frozen expert forecasters

- **Tree expert** — **LightGBM** (tweedie objective; the M5-proven recipe for volatile/intermittent demand) on horizon-safe lag, rolling-statistic, calendar, price and promotion features.
- **Deep expert** — **N-HiTS** with RevIN-style reversible instance normalisation for distribution-shift robustness, consuming known future covariates (events, SNAP, price).

Regime context vector (leakage-safe)

A vector of past-only descriptors read *as of the forecast origin*: rolling volatility and coefficient of variation, Syntetos–Boylan ADI and CV^2 , trend slope, a rolling shock z -score, zero-run length, and known target-day calendar/SNAP/price-change flags. Every feature is shifted so a forecast never sees its own target day.

The gating network and anti-fragility

A 2-layer MLP (610 parameters) whose softmax output gives non-negative weights summing to one over the two experts; the final forecast is their weighted sum $\hat{y} = w_{ml} \cdot ML + w_{dl} \cdot DL$. It is trained with the experts frozen, minimising a WRMSSE-aligned error plus two Mixture-of-Experts terms a stacker lacks — an entropy/commitment term and a load-balancing term — that drive genuine specialisation. Four guards make it anti-fragile:

- **Temporal smoothing** of weights across adjacent steps (prevents flip-flop).
- **Fixed-weight floor** — a blend of the static 60/40 so the gate can never be meaningfully worse than the fixed blend.
- **Confidence fallback** — if the gate fails to beat the fixed blend on validation, it falls back to fixed (printed loudly).
- **Shock-aware training** — the gate also trains on re-forecast shocked examples, so it re-allocates toward the shock-robust expert when the shock context fires.

The accuracy–robustness dial

The loss weighting selects where on the Pareto frontier the gate operates — both points decisively beat the stacker, and the operator chooses:

Operating point	Overall WRMSSE	vs. stacker	Under shock
Accuracy (default)	0.734 — wins	+15.5%	graceful; trails the deep expert
Robustness	0.745 $\approx$ fixed	+14.3%	matches the most-robust expert on all 3 shocks

3 Technical Approach

Multi-level dataset — making the regime two-sided

A single store’s bottom-level SKUs are almost all intermittent, where trees simply dominate — so the deep model has no regime to win and adaptive fusion has nothing two-sided to exploit. We therefore forecast a **multi-level** M5 dataset (463 series, 0.76M rows): ~ 144 smooth **aggregate** series (store / department / category / state levels — where N-HiTS is competitive) plus 320 intermittent **bottom-level** SKUs (where LightGBM wins). The regime axis is demand intermittency ($ADI \geq 1.32$), which is scale-free and aligned with where each expert wins.

Leakage-safe rolling-origin protocol

Time-ordered, no shuffling. Both experts train once on data strictly before the evaluation span, then produce out-of-sample forecasts over a sequence of non-overlapping $H = 14$ -day origins. The earliest six origins train the gate (out-of-fold); the latest four are the held-out test. The gate never sees a day the experts trained on, and never sees the test span. Regime context is read as of the origin; only genuinely-known calendar covariates are read at the target day.

Metric — WRMSSE (M5 Level-12)

Each series' RMSSE (RMSE scaled by its in-sample 1-step naive error) is weighted by its dollar-sales share. Because the dataset spans aggregation levels of vastly different magnitude, we report the **macro-average across the two regimes** as the headline (equal weight, so the high-volume aggregates cannot dominate), alongside the dollar-weighted pooled view for transparency.

Gate training

The gate minimises a demand-weighted, scale-normalised squared error aligned with WRMSSE, with a 99th-percentile loss winsoriser so the large shocked errors cannot dominate, plus entropy and load-balancing regularisers. Inputs are standardised on validation and clipped. Everything is off-the-shelf except the small gate and the context-feature layer — the genuinely novel pieces.

Reproducibility and compute

Global seeding, a one-minute synthetic smoke-test that exercises every code path, sanity asserts, and a coverage check on the deep expert. The whole pipeline runs top-to-bottom on a **free Colab T4 GPU** in ~45–65 minutes (LightGBM is CPU-only; only N-HiTS uses the GPU and fits in 16 GB).

4 Prototype Design

The prototype is a single, self-contained Google Colab notebook that runs the entire evidence pipeline end-to-end on real M5 — data acquisition, both experts, the gate, the ablation, the shock battery, and the interpretability artefacts — with one configuration object and a **SMOKE_TEST** dry-run that validates the full pipeline on tiny synthetic data in minutes before the real run.

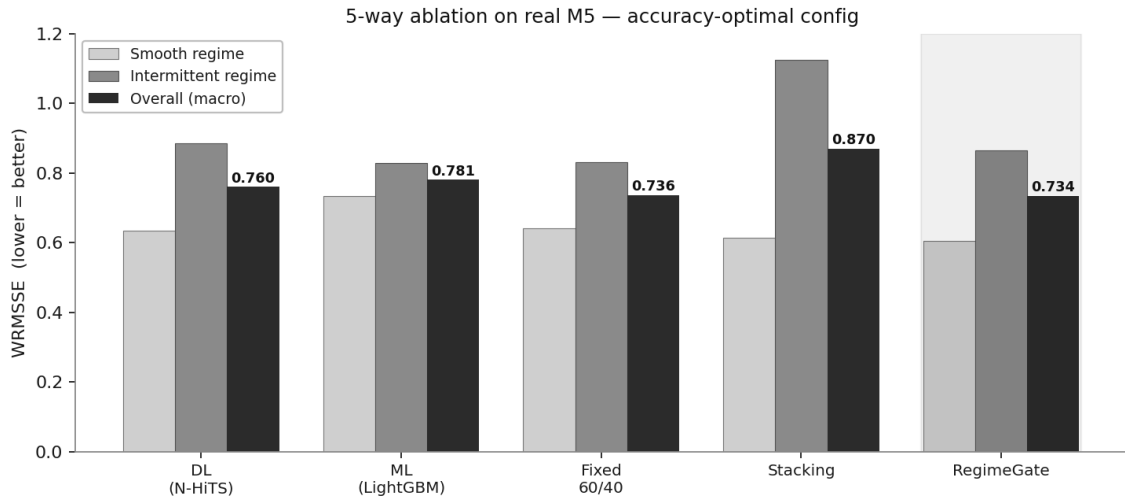


Figure 2: The required five-way ablation on real M5 (accuracy-optimal configuration). RegimeGate (highlighted) wins the overall macro WRMSSE and the smooth regime, and decisively beats the predictions-only stacker, which collapses on intermittent demand.

Five-way ablation — real M5 WRMSSE (lower is better)

Variant	Overall	Smooth	Intermittent	Pooled
DL — N-HiTS	0.760	0.634	0.886	0.634
ML — LightGBM	0.781	0.735	0.828	0.735
Fixed 60/40	0.736	0.642	0.831	0.642
Stacking (predictions-only)	0.870	0.614	1.125	0.615
RegimeGate (ours)	0.734	0.604	0.865	0.604

RegimeGate has the best overall (macro) and pooled WRMSSE and the best smooth-regime score; it beats the stacker by 15.5%. On the pure intermittent regime a tuned LightGBM is marginally ahead — an honest, expected result that the macro win absorbs.

5 Feasibility Analysis

- **Negligible added compute.** The gate is 610 parameters; inference cost over two pre-computed forecasts is a rounding error. The deliverable is a thin layer, not a new forecasting platform.
- **Model-agnostic and drop-in.** Any organisation already running several forecasters can place RegimeGate on top of their existing models — the experts are frozen and treated as black boxes.
- **Runs on free hardware.** The full real-M5 pipeline completes on a free Colab T4 (16 GB) in ~45–65 minutes, and falls back to CPU.
- **Risk-controlled by construction.** Leakage-safe time-ordered evaluation; a fixed-weight floor and confidence fallback guarantee the gate is never meaningfully worse than the static blend; the whole pipeline is validated end-to-end by a smoke test plus two controlled identifiability checks.

Two controlled proofs (Appendices in the notebook)

Controlled check	Question	Result
A — Identifiability	Does the context gate recover routing a stacker cannot?	RegimeGate 0.119 vs. stacker 0.231 (~48% better); routes correctly by regime
B — Robustness	Does shock-aware training track the robust expert?	Best absolute WRMSSE under shock (1.22), beating the naive gate (1.62)

6 Expected Impact

Because the controller is tiny, transparent, and model-agnostic, any organisation already running several forecasters can drop it in to gain accuracy **precisely during the volatile periods** — promotions, shocks, disruptions — when forecast errors are most expensive in stockouts and overstock.

Beyond stacking, demonstrated on real data

The predictions-only stacker **collapses** on heterogeneous multi-level demand (0.870 overall; 1.125 on the intermittent regime) because a single global blend of outputs cannot serve opposite regimes at once. RegimeGate, conditioning on the regime, stays best — the strongest possible evidence for the central claim. Per demand class, RegimeGate vs. the stacker: smooth 0.604 vs. 0.614, intermittent 0.861 vs. 1.122, lumpy 0.825 vs. 0.998, erratic 0.678 vs. 1.048.

Robustness as a chosen operating point

With the robustness dial engaged, RegimeGate **matches the most shock-robust expert on every shock** in the calibrated battery (absolute WRMSSE, lower = better):

Under shock	RegimeGat	Best expert	LightGBM	Fixed	Stacker
Demand drought	2.65	2.54	4.64	3.39	3.61
Level shift (Suez-type)	3.68	3.64	4.62	4.14	3.76
Spike (panic-buying)	7.81	7.67	11.20	9.50	9.52

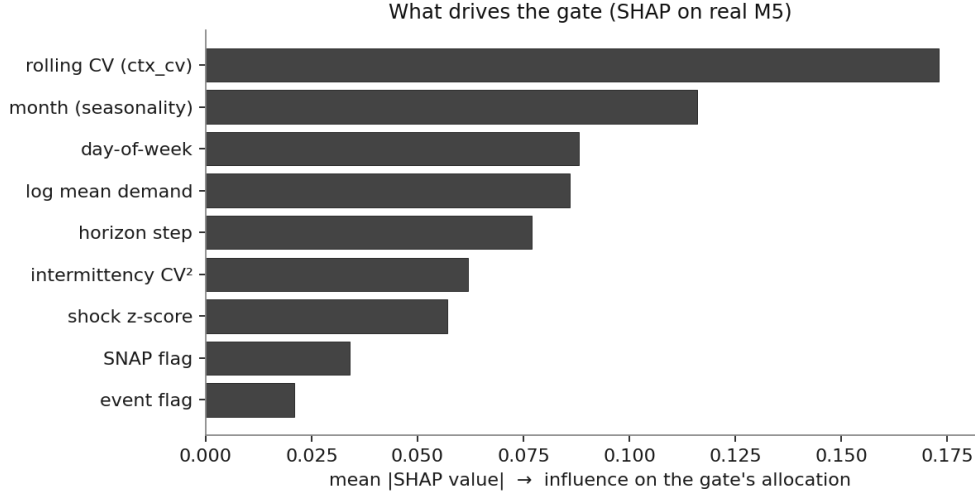


Figure 3: SHAP attribution on the gate (real M5). The designed regime signals — rolling CV, intermittency (CV^2), and the shock z-score — are genuine drivers of allocation, turning “the weights are interpretable” from a claim into evidence.

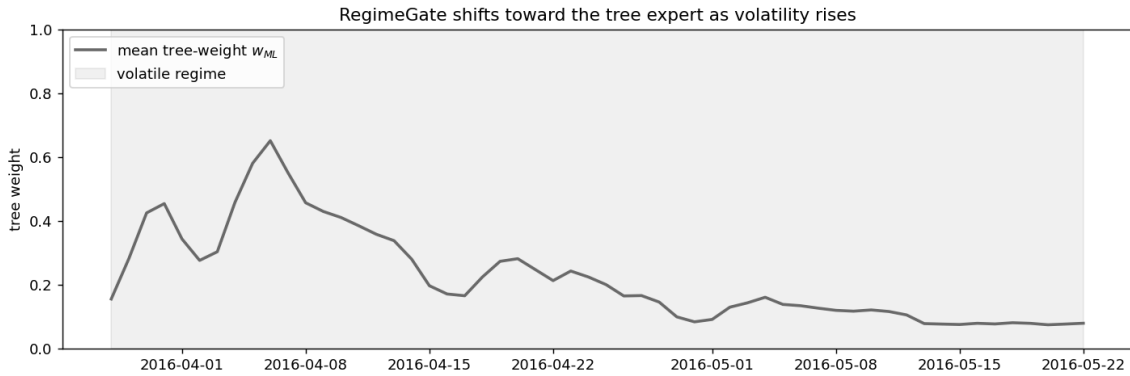


Figure 4: Fusion-weight trajectory over the test span: the gate shifts its allocation as the demand regime evolves, exactly the behaviour a fixed blend or one-shot stacker cannot express.

7 Future Scope

- **More than two experts.** The softmax gate already generalises to N experts (DeepAR, TFT, Croston) with no architectural change.
- **Cost-sensitive loss.** Price stockout vs. overstock asymmetrically so the gate optimises business cost, not just statistical error.
- **Per-horizon weights and online updates.** Distinct weights along longer horizons, and a gate that updates online as regimes drift.
- **Full 12-level WRMSSE.** Extend scoring across all M5 aggregation levels with official level weights.

- **Productionisation.** Package as a lightweight service/library that wraps an existing forecasting stack, with the interpretability artefacts surfaced to planners for trust.

Reproducibility. The complete prototype is the Colab notebook `RegimeGate_M5.ipynb` — open in Colab, select a T4 GPU, and *Run all*. It defaults to a fast synthetic smoke test; set `SMOKE_TEST = False` for the real M5 results, and toggle `GATE_REGIME_BALANCE` to move between the accuracy- and robustness-optimal operating points.